
Topic 7

Thinking and language

Overview

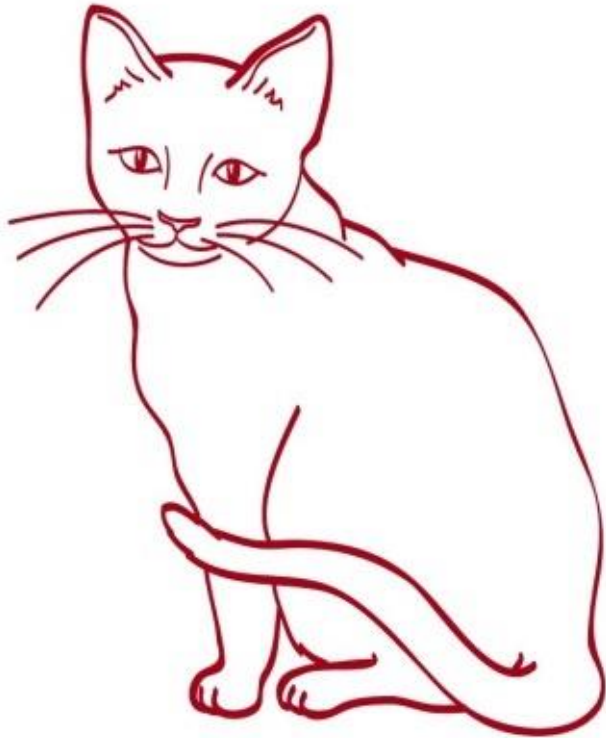
- Mental representations
- Decision making
- Problem solving
- Language

Mental representations

- Components of knowledge:
 - Mental representations
 - Analogical representations
 - Symbolic representations

Mental representations

(A) Analogical representation



(B) Symbolic representation

cat

猫

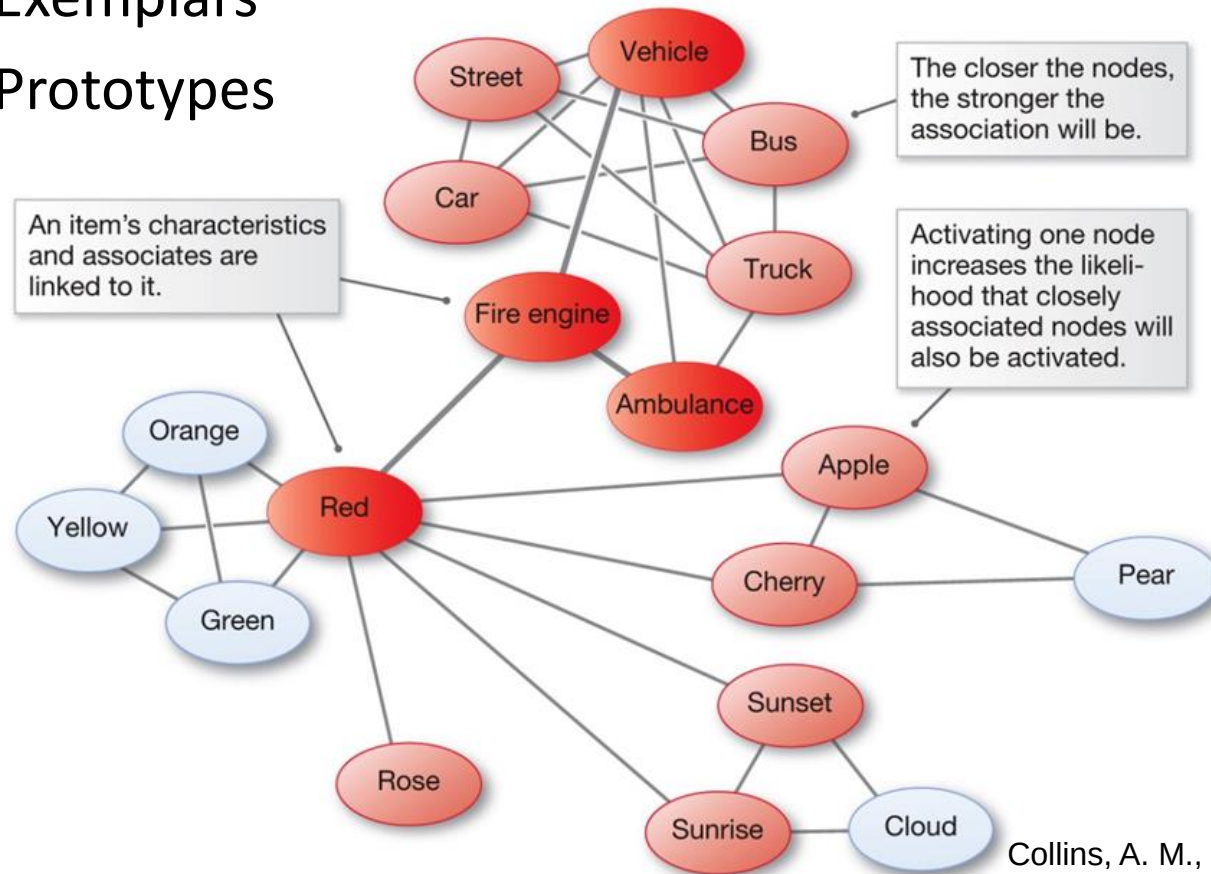
Mental representations

- Analogical representations
 - Analogical representations are dissimilar from pictures
 - What did you see?
 - Which is farther east, San Diego, California, or Reno, Nevada?



Mental representations

- Symbolic representations
 - How do we represent concepts in our mind such that we can call up information quickly?
 - Exemplars
 - Prototypes



Categorization

- **Prototype theory** (Rosch, 1973):
 - Category membership depends on the distance (similarity) to a *prototype*.
 - **Prototype**
 - An “average” representation of the most common members of a category (not a member itself)
 - Membership can vary in strength, depending on representational similarity.



FIGURE 9.4

Three real birds—a sparrow, a robin and a jay—and a “prototype” bird that is the average representation of these three category members.

Categorization

- **Exemplar theory** (Nosofsky, 1986):
 - Category membership depends on similarity to *exemplars*.

Uses real examples to compare with



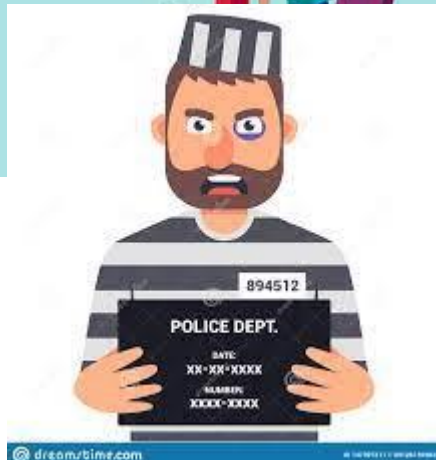
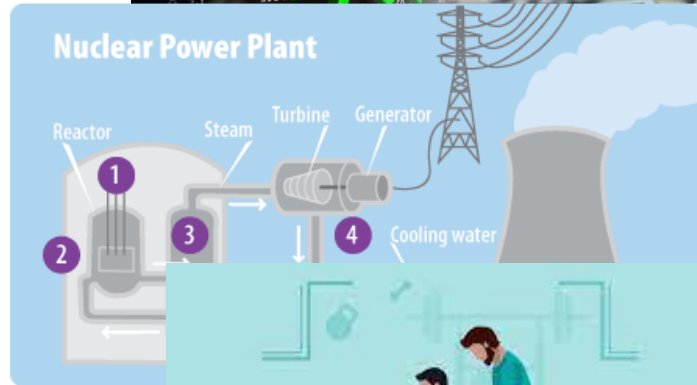
Similar to **many** earlier
examples that you have seen:
typical



Similar to **few** examples
that you have seen:
atypical

Overview

- Mental representations
- **Decision making**
- Problem solving
- Language



- **Judgement**
 - Deciding on the likelihood of various events using incomplete information
- **Decision making**
 - Selecting one option among several possibilities

Traditional decision research

- **Rational choice:**
 - A single “correct” choice after a cognitive evaluation
- Behavioral violations of “rational choice models”

Decision making

- **Normative decision theories** define how people should make decisions.
 - **Expected utility theory**
 - Decisions should be made to maximize the outcome.
 - This should depend on
 - the values of the alternative options
 - the probabilities associated with the alternative options
 - e.g., a lottery ticket costs € 10,- and $p(\text{winning € 1.000.000,-}) = 10^{-6}$
 - $EV(\text{buying}) = ((1-10^{-6}) * -€ 10,-) + (10^{-6} * € 1.000.000,-) = -€ 9,-$
 - $EV(\text{not buying}) = (1 * € 0,-) + (0 * € 1.000.000,-) = € 0,-$

Decision making

- **Normative decision theories** define how people should make decisions.
 - **Expected utility theory**
 - Decisions should be made to maximize the outcome.
 - This should depend on
 - the values of the alternative options
 - the probabilities associated with the alternative options
- **Descriptive decision theories** indicate how people actually make choices.

Decision making

- **Heuristics:**
 - shortcuts (rules of thumb or informal guidelines) used to reduce the amount of thinking that is needed to make decisions
 - Heuristic thinking occurs often unconsciously.
 - Heuristic thinking allows us to decide quickly.
 - Heuristics can also result in errors or faulty decisions.

Decision making

- **Anchoring:**

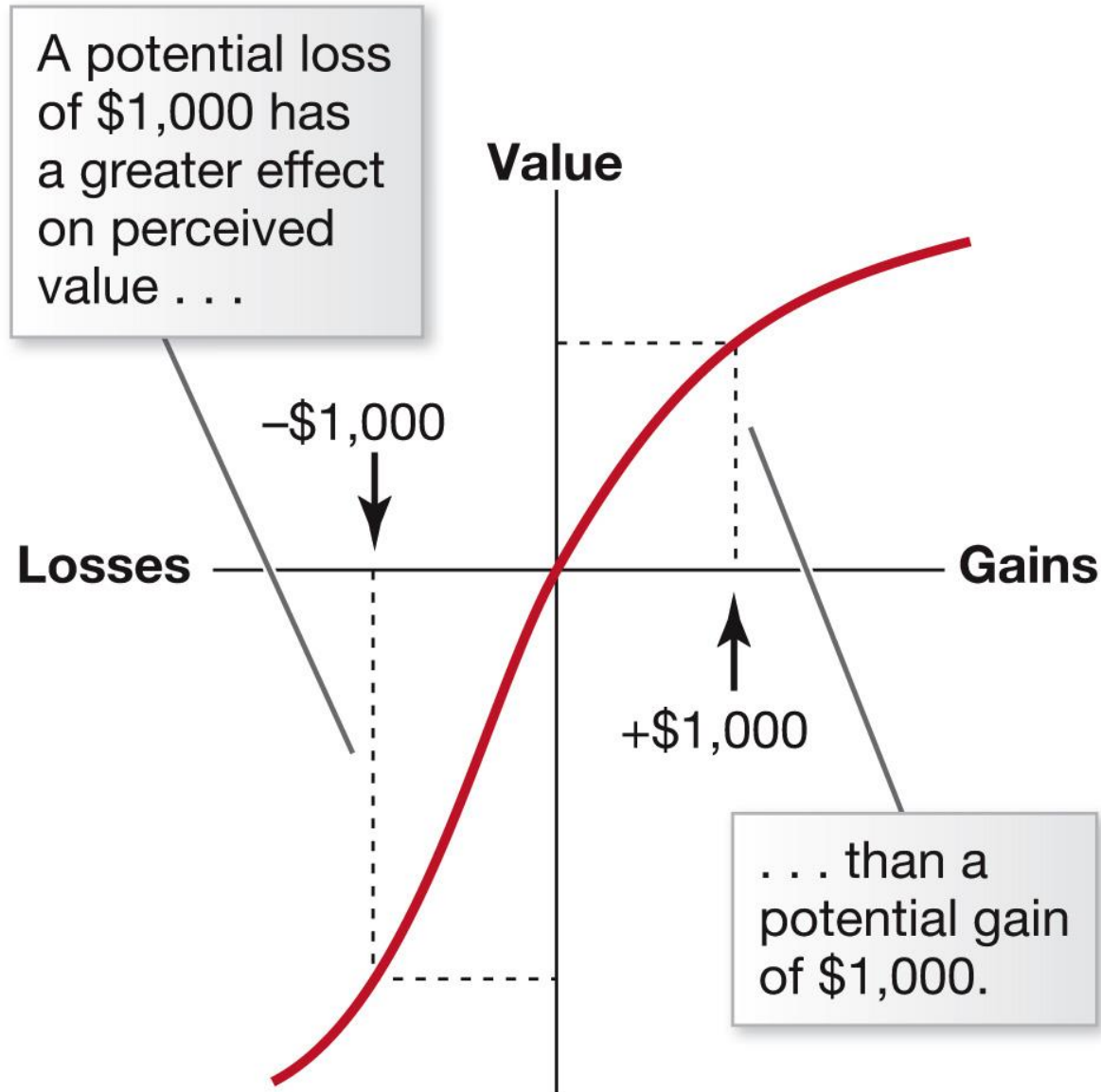
- the tendency, in making judgments, to rely on the first piece of information encountered

- **Framing:**

- in decision making, the tendency to emphasize the potential losses or potential gains from at least one alternative



Decision making



- **Framing effect**

- Asian disease problem (Tversky & Kahneman, 1981)
 - 600 people in danger

	Positive Frame	Negative Frame
Treatment A	72% Saves 200 lives	400 people will die
Treatment B	33% chance of saving all 600 people 66% possibility of saving no one	%33 chance that no people will die 66% probability that all 600 people will die

78%

Gains

Most participants fail to make the best choice.

Losses

Would you prefer:

☐ **100%** chance of gaining
800€

OR

☐ **85%** chance of gaining
1000€
(15% chance gaining nothing)

Risk aversion

Preferring a sure gain to a risky gain

Would you prefer:

☐ **100%** chance of losing
800€

OR

☐ **85%** chance of losing
1000€
(15% chance losing nothing)

Loss aversion

More sensitivity to potential losses than potential gains.

Decision making

- **Availability heuristic:**
 - deciding based on the answer that most easily comes to mind
 - e.g., death by a car accident vs brain hemorrhage
- **Representativeness heuristic:**
 - deciding based on how typical an individual, object, or event belongs in a certain category

Decision making

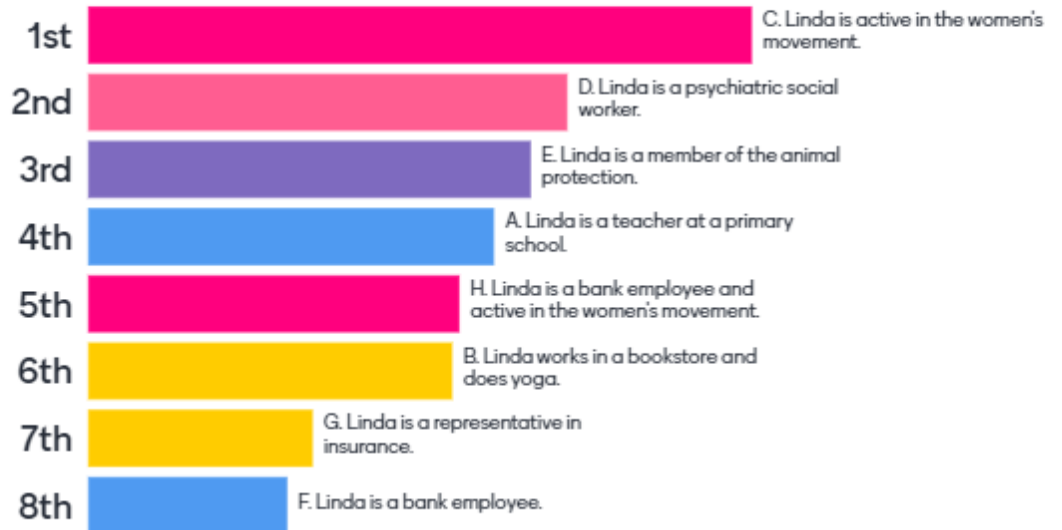
Linda is 56 years old, single, outgoing and smart. She has studied philosophy and has always been heavily involved in issues concerning human rights, social equality and discrimination. In the 1980s, she has often participated in demonstrations against nuclear weapons.

- A. Linda is a teacher at a primary school.
- B. Linda works in a bookstore and does yoga.
- C. Linda is active in the women's movement.
- D. Linda is a psychiatric social worker.
- E. Linda is a member of the animal protection.
- F. Linda is a bank employee.
- G. Linda is a representative in insurance.
- H. Linda is a bank employee and active in the women's movement.



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Which is more likely about Linda?



Decision making

8 coin tosses

Which outcome is more probable?

(1)

head

head

tails

head

tails

tails

tails

head

(2)

head

head

head

head

head

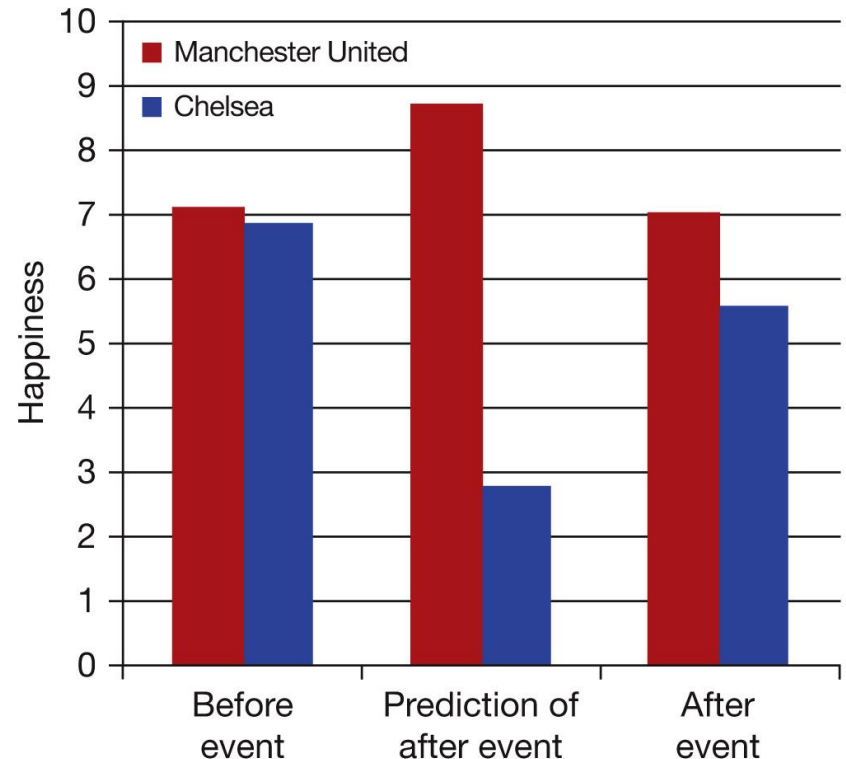
head

head

head

Decision making

- **Affective forecasting heuristic:**
 - the tendency for people to overestimate how events will make them feel in the future



Decision making

- **Affective forecasting heuristic:**
 - the tendency for people to overestimate how events will make them feel in the future
- **The paradox of choice:**
 - when too many options are available, especially when all of them are attractive, people experience conflict and indecision



Decision making

- A good strategy is to satisfice rather than optimize your decisions

Scenario A

Suppose you are considering buying **headphones** and have not yet decided which model to buy. You pass a store that is having a 1-day clearance sale. It offers a popular Sony player for just \$99, well below the list price. Do you

A. buy the Sony player? 66%

B. wait until you learn more about the various models? 34%

Scenario B

Suppose you are considering buying **headphones** and have not yet decided which model to buy. You pass a store that is having a 1-day clearance sale. It offers a popular Sony player for just \$99 and a top-of-the-line Aiwa player for just \$169, both well below the list price. Do you

A. buy the Sony player? 27%

B. buy the Aiwa player? 27%

C. wait until you learn more about the various models? 46%

Decision Making

- Satisficers vs. Maximizers

Overview

- Mental representations
- Decision making
- **Problem solving**
- Language

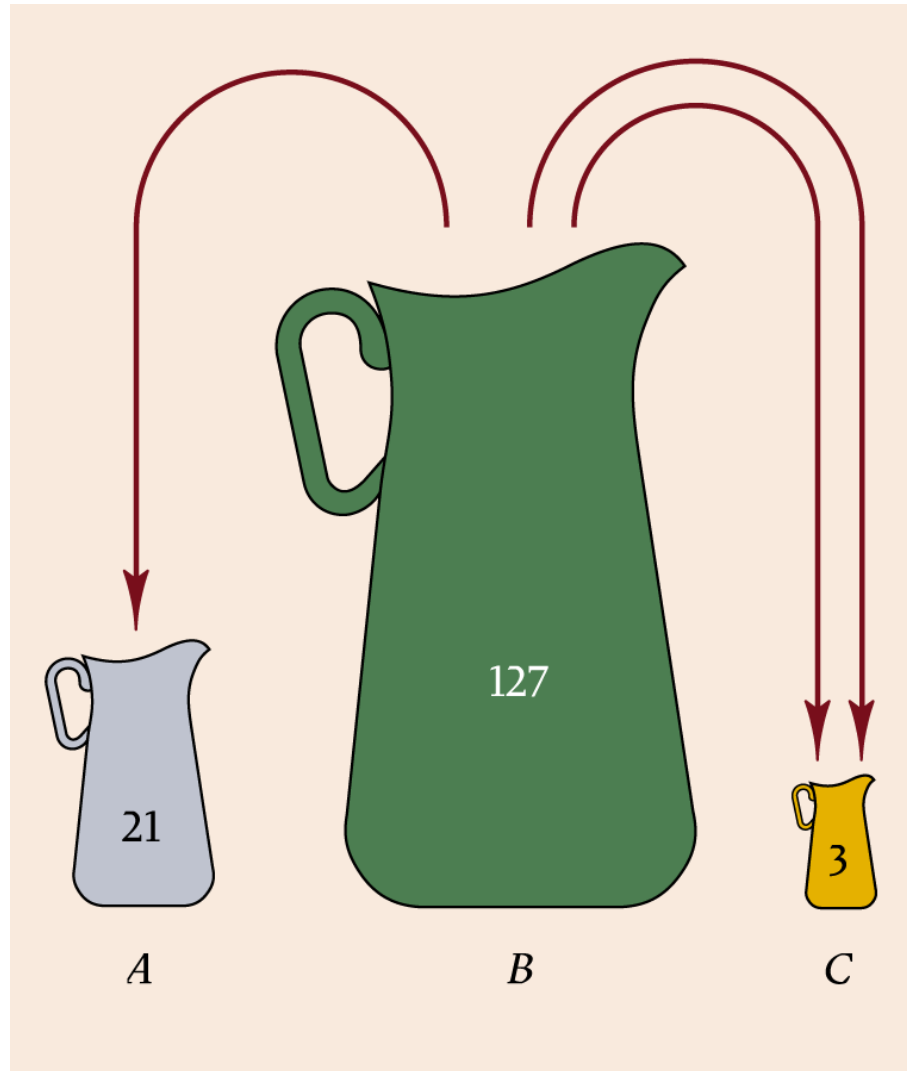
Problem solving

- Thinking process in which we move from an initial state to a goal state →
 - goal-directed

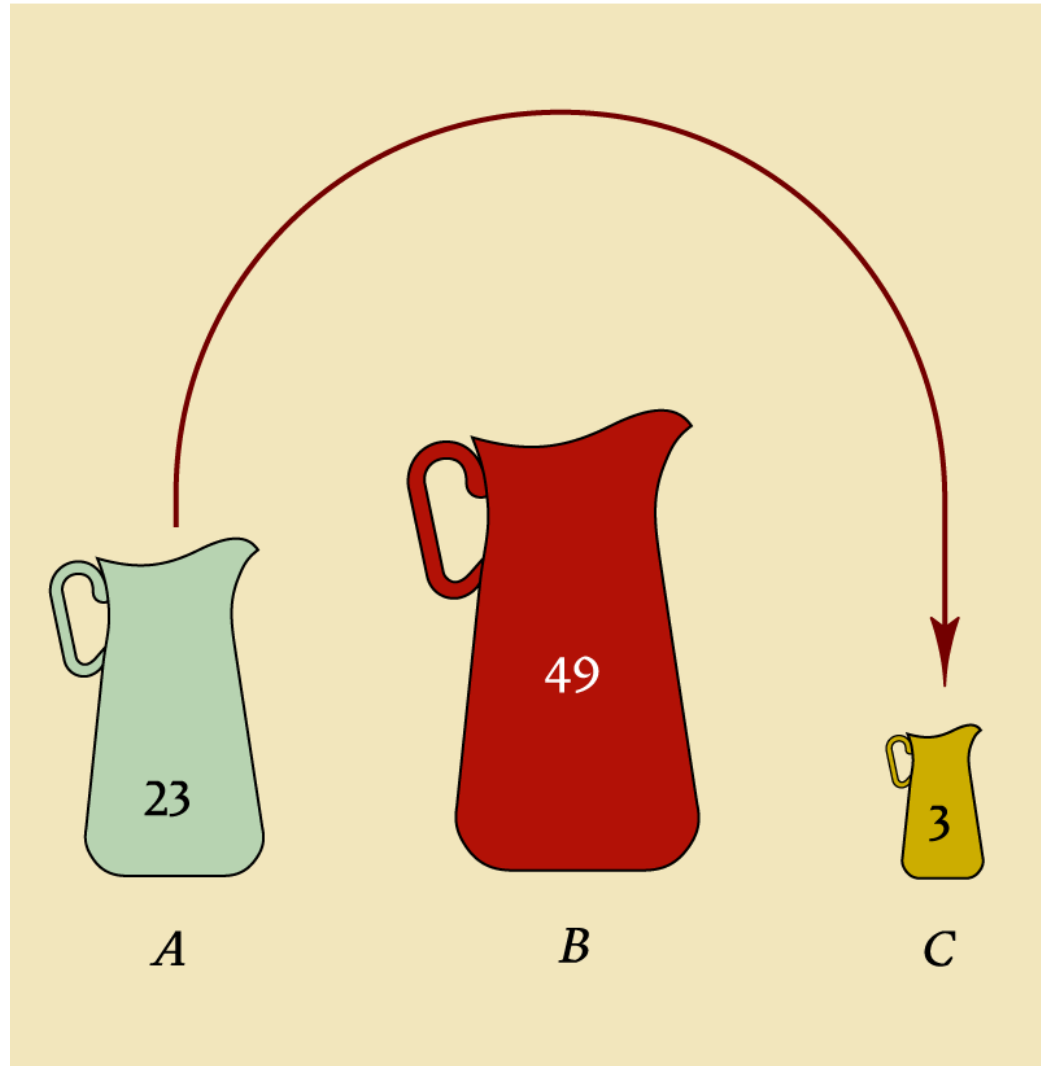
Problem solving

- Obstacles to problem solving
 - Complexity
 - Mental set
 - Instruction / practice

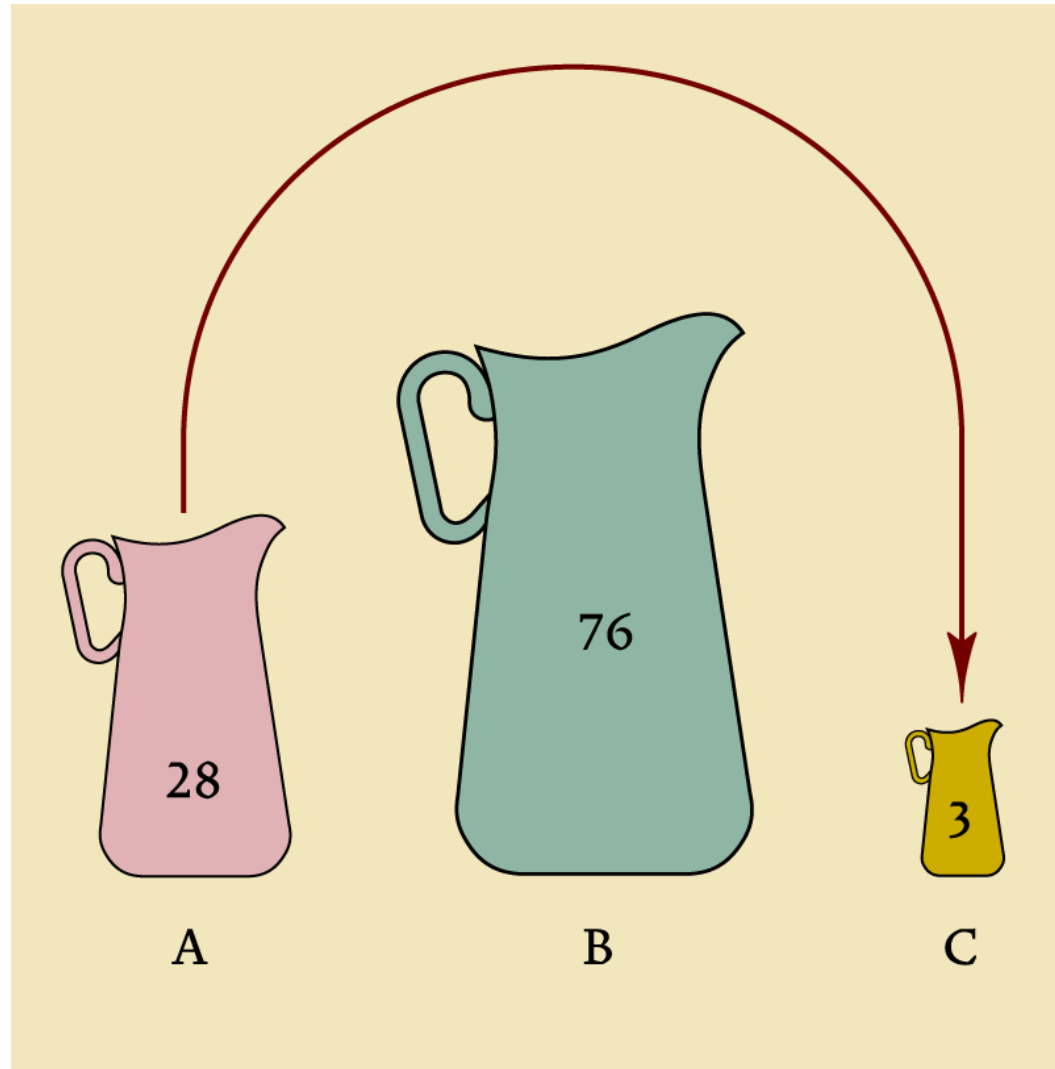
Problem solving



Problem solving



Problem solving



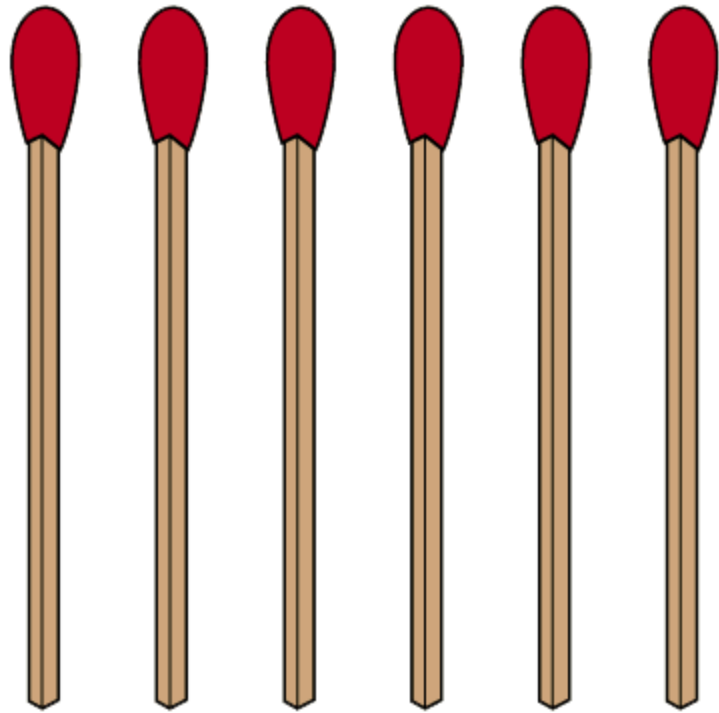
Problem solving

- Obstacles to problem solving
 - Complexity
 - Mental set
 - Instruction / practice
 - Perceptual organization of the problem

Problem solving



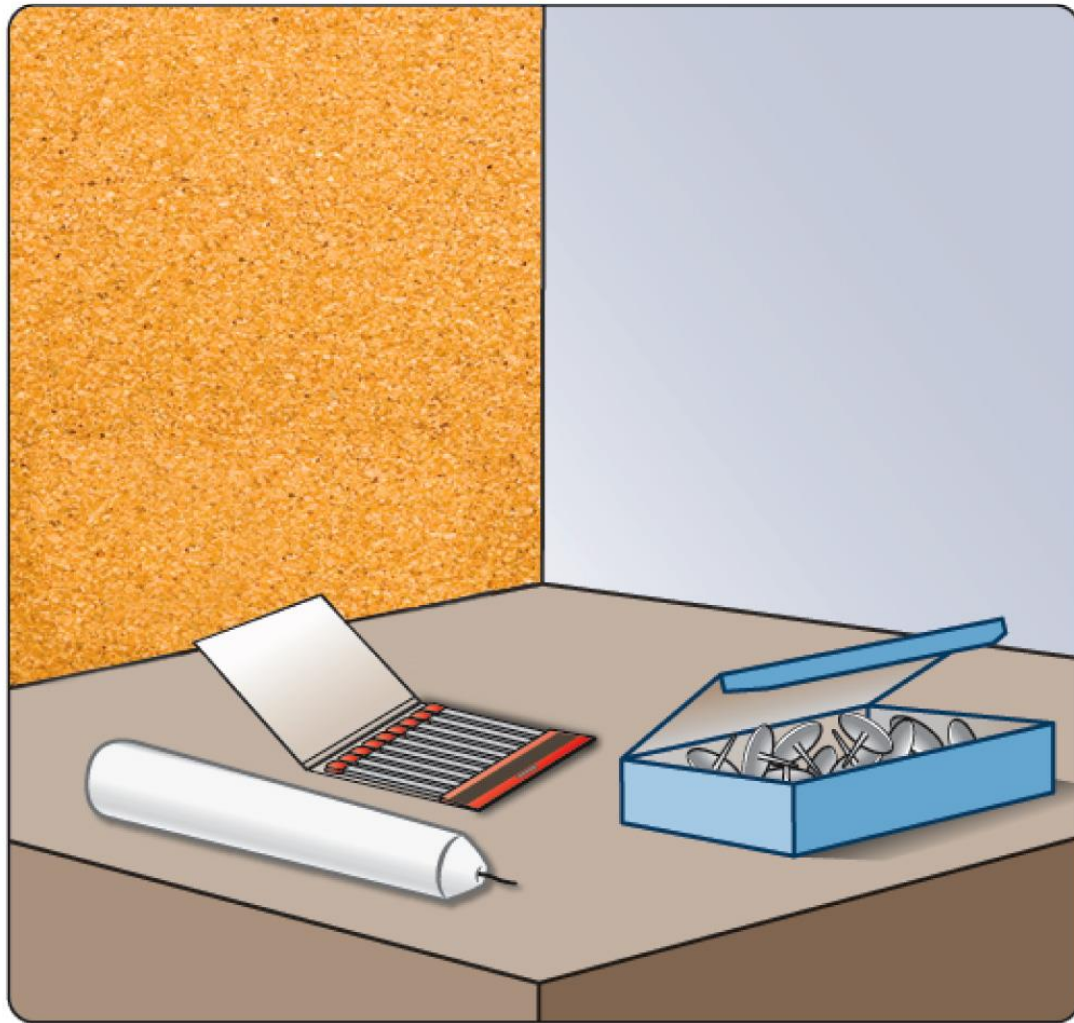
Problem solving



Problem solving

- Obstacles to problem solving
 - Complexity
 - Mental set
 - Instruction / practice
 - Perceptual organization of the problem
 - Functional fixedness:
 - having fixed ideas about the typical functions of objects

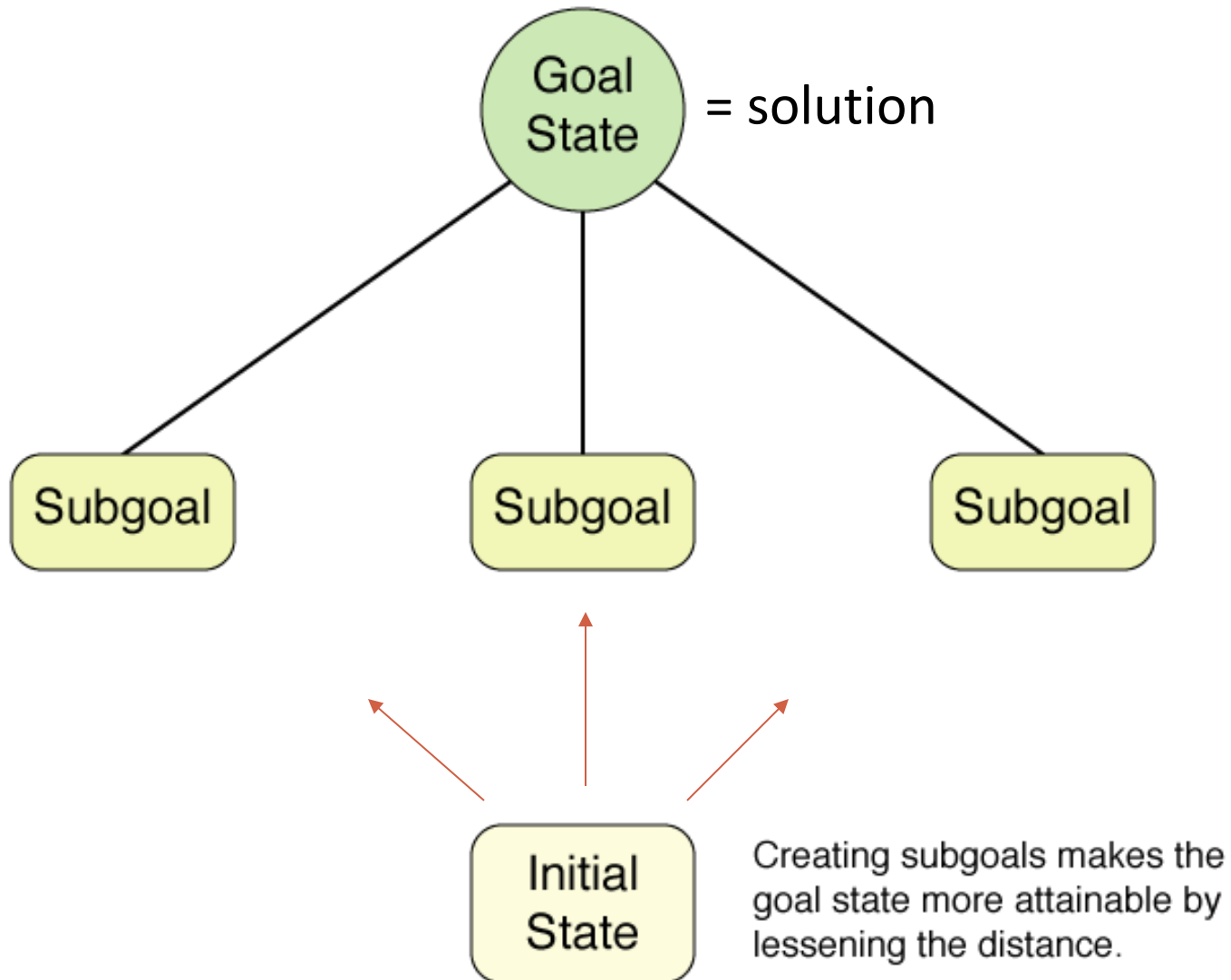
Problem solving



Problem solving

- Overcoming obstacles to solutions
 - Understanding → formulate subgoals

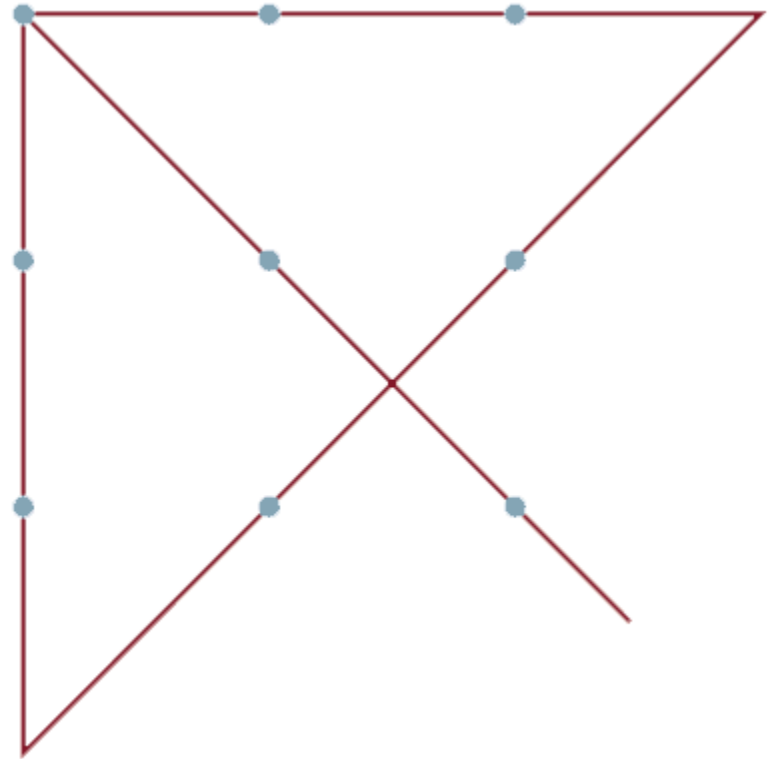
Problem solving



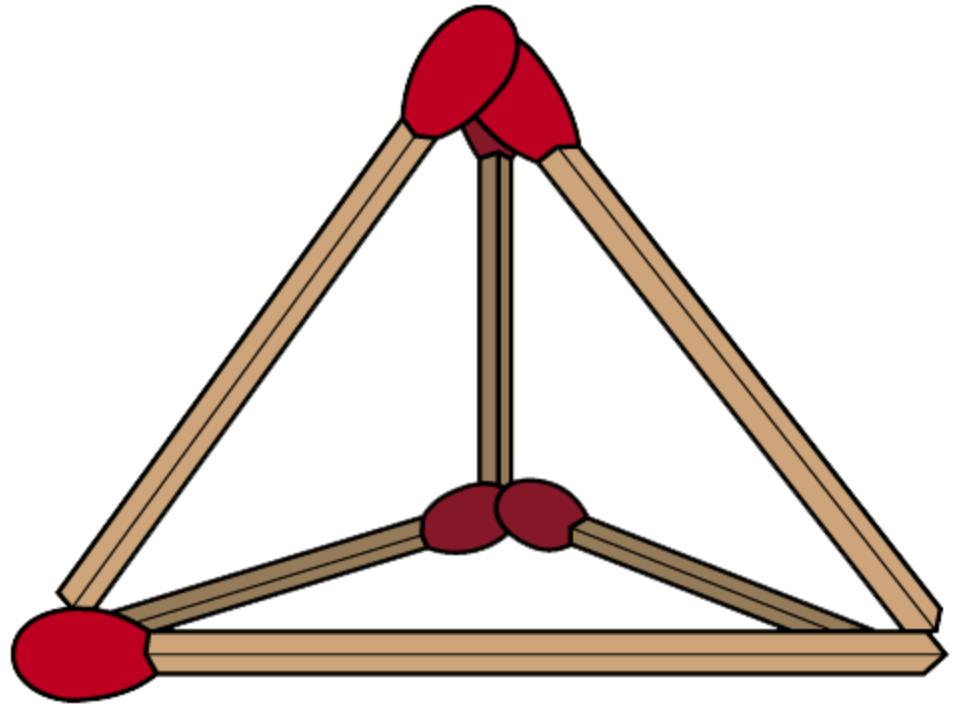
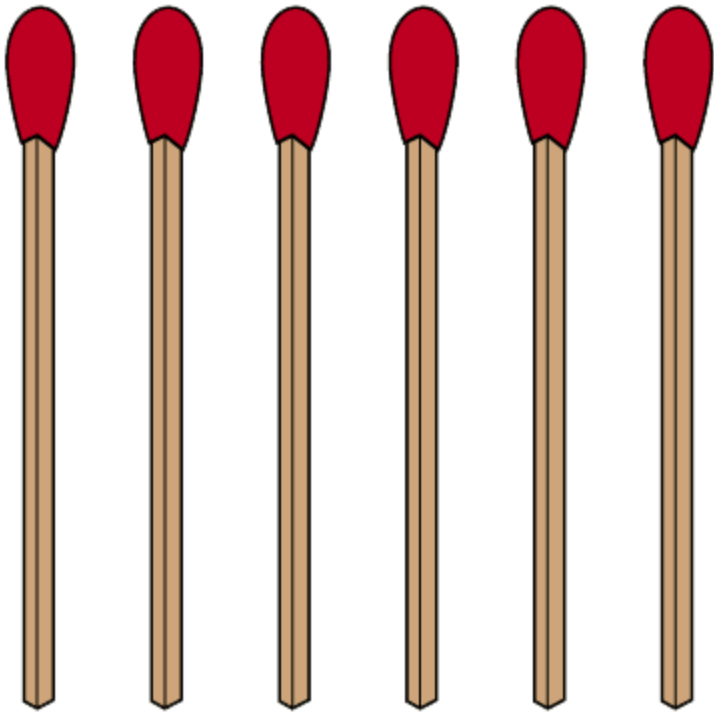
Problem solving

- Overcoming obstacles to solutions
 - Understanding → formulate subgoals
 - Restructuring

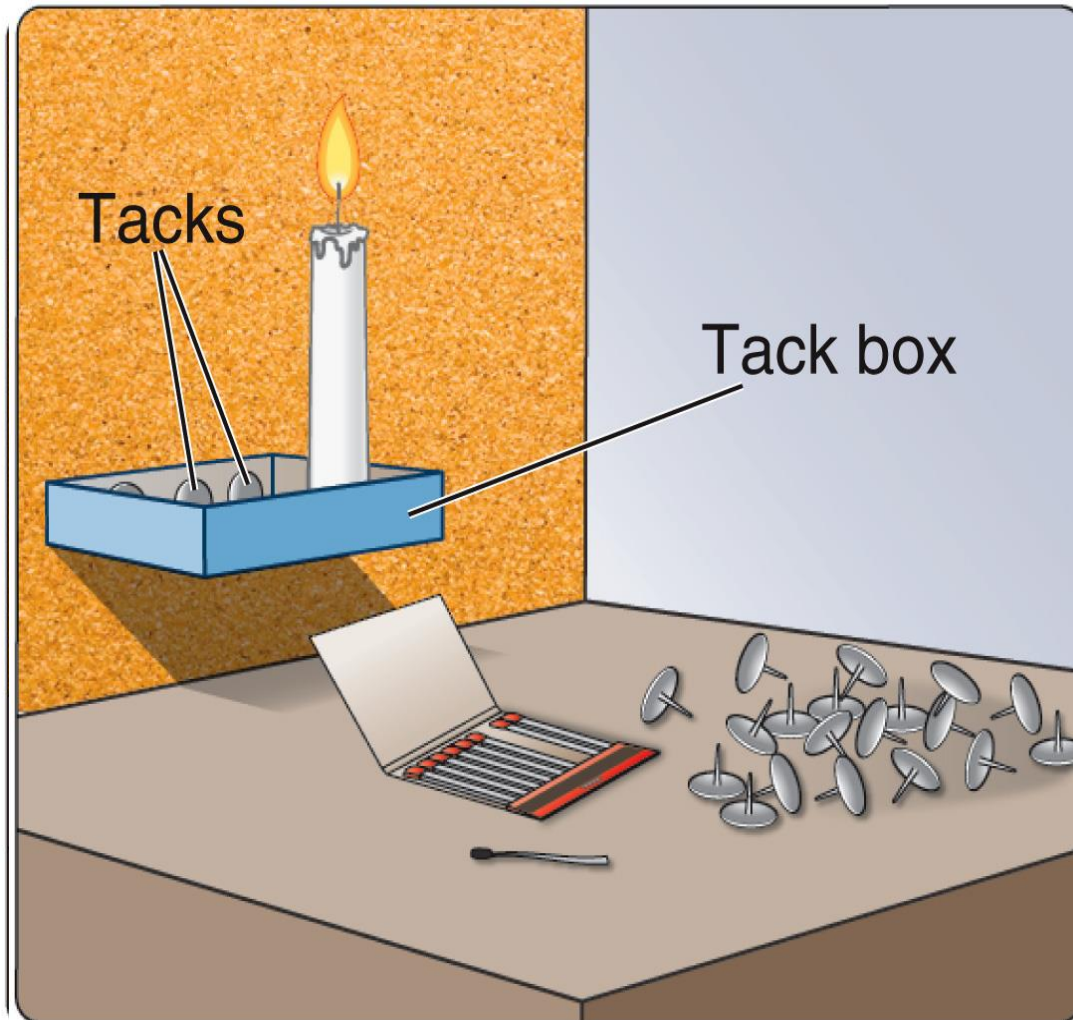
Problem solving



Problem solving

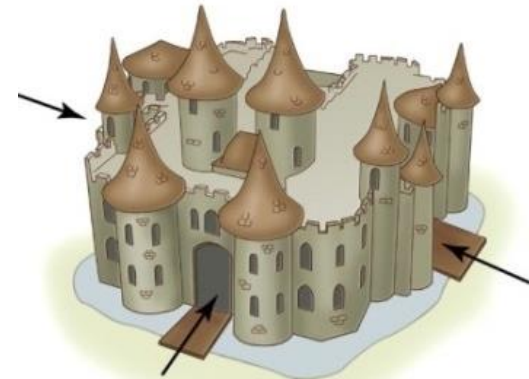
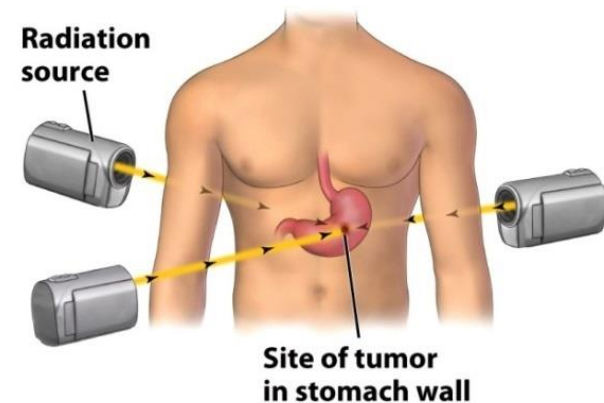


Problem solving



Problem solving

- Overcoming obstacles to solutions
 - Understanding → formulate subgoals
 - Restructuring
 - Use an algorithm
 - Use an analogy
 - Wait for a sudden insight



Problem solving

(a)



(b)



Overview

- Mental representations
- Decision making
- Problem solving
- **Language**

Language

- **Language:** a system of communication using sounds and symbols according to grammatical rules
 - Syntax: rules
 - Semantics: meanings

Language

Sentence

Stephanie kissed the crying boy

Phrases

Stephanie

kissed the crying boy

Words or
morphemes

Stephanie

kiss

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cry

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boy

Phonemes

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Mercedes

Language

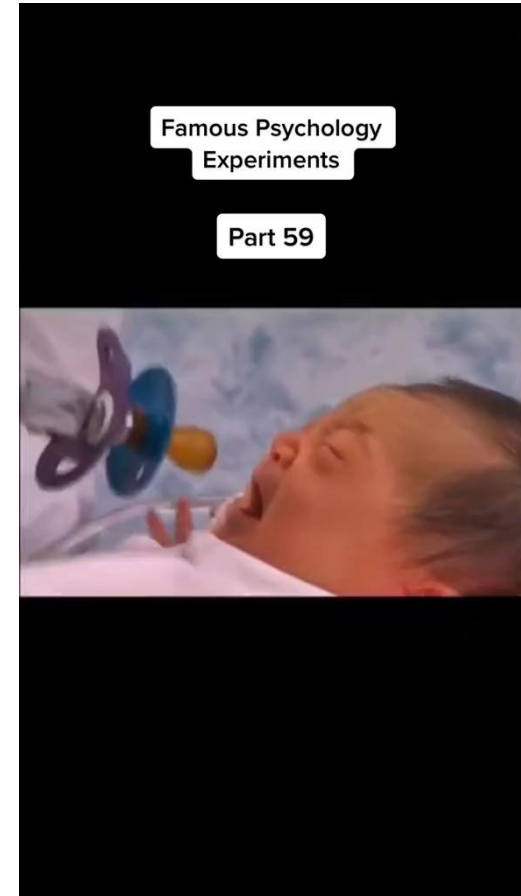
Major Consonants and Vowels of English and Their Phonetic Symbols

CONSONANTS				VOWELS	
p	<i>pull</i>	s	<i>sip</i>	i	<i>heed</i>
b	<i>bull</i>	z	<i>zip</i>	ɪ	<i>hid</i>
m	<i>man</i>	r	<i>rip</i>	e	<i>bait</i>
w	<i>will</i>	ʃ	<i>should</i>	ɛ	<i>head</i>
f	<i>fill</i>	ʒ	<i>pleasure</i>	æ	<i>had</i>
v	<i>vet</i>	č	<i>chop</i>	u	<i>who'd</i>
θ	<i>thigh</i>	ĵ	<i>gyp</i>	ʊ	<i>put</i>
ð	<i>that</i>	y	<i>yip</i>	ʌ	<i>but</i>
t	<i>tie</i>	k	<i>kale</i>	o	<i>boat</i>
d	<i>die</i>	g	<i>gale</i>	ɔ	<i>bought</i>
n	<i>near</i>	h	<i>hail</i>	a	<i>hot</i>
l	<i>lear</i>	ŋ	<i>sing</i>	ə	<i>sofa</i>
				ɪ	<i>many</i>

There are other American English phonemes in addition to those shown here, and specific symbols may vary depending on the source.

Language

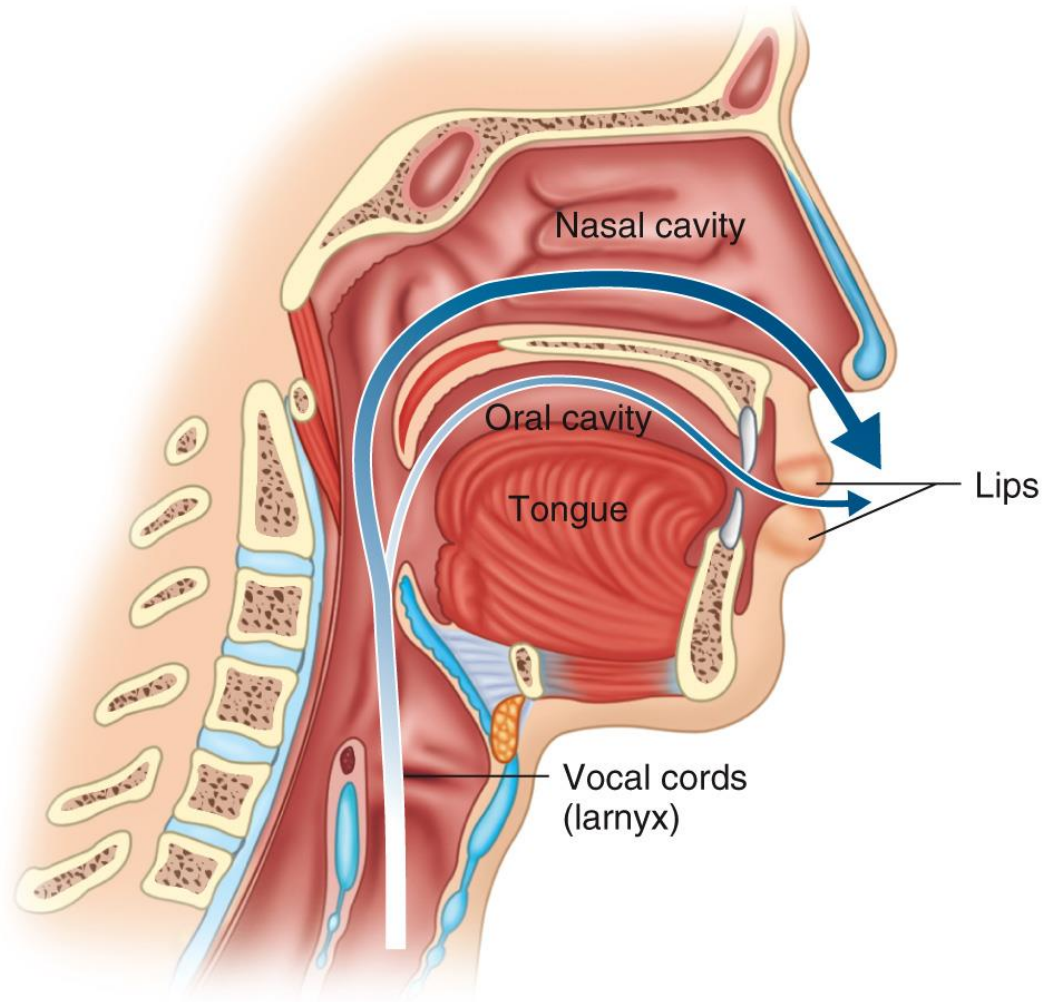
- As the brain develops, so does the ability to speak and form sentences.
 - Development occurs in an orderly way
 - Listening preferences in newborns are influenced by the language or languages spoken by mothers during pregnancy.
 - Newborns can discriminate all the phonemes but lose this ability after 6 months.



Language

- The human vocal tract has the capacity to make many more sounds than any language uses.
 - Languages differ from one another not only by the words that are used but also by the number of phonemes and the patterns of morphemes.

Language



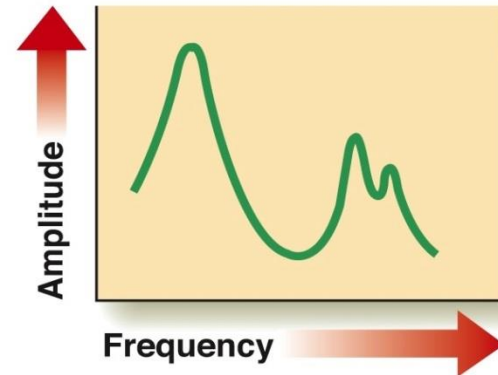
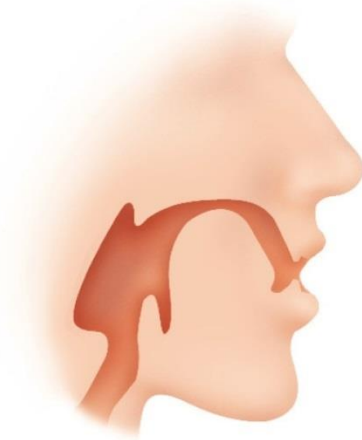
Language

Phoneme
symbol

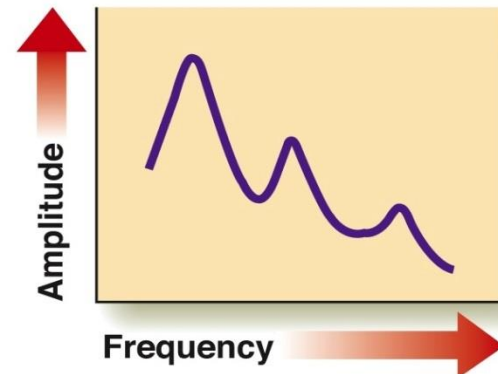
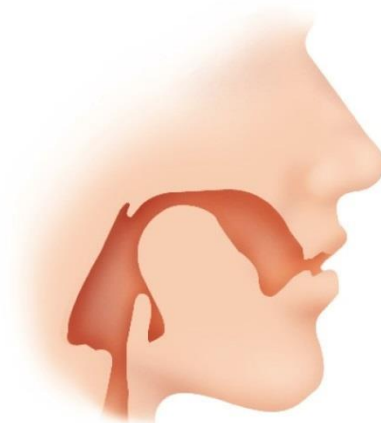
Outline of vocal tract
traced from x-ray
picture of mouth

Pressure
changes

/ I /



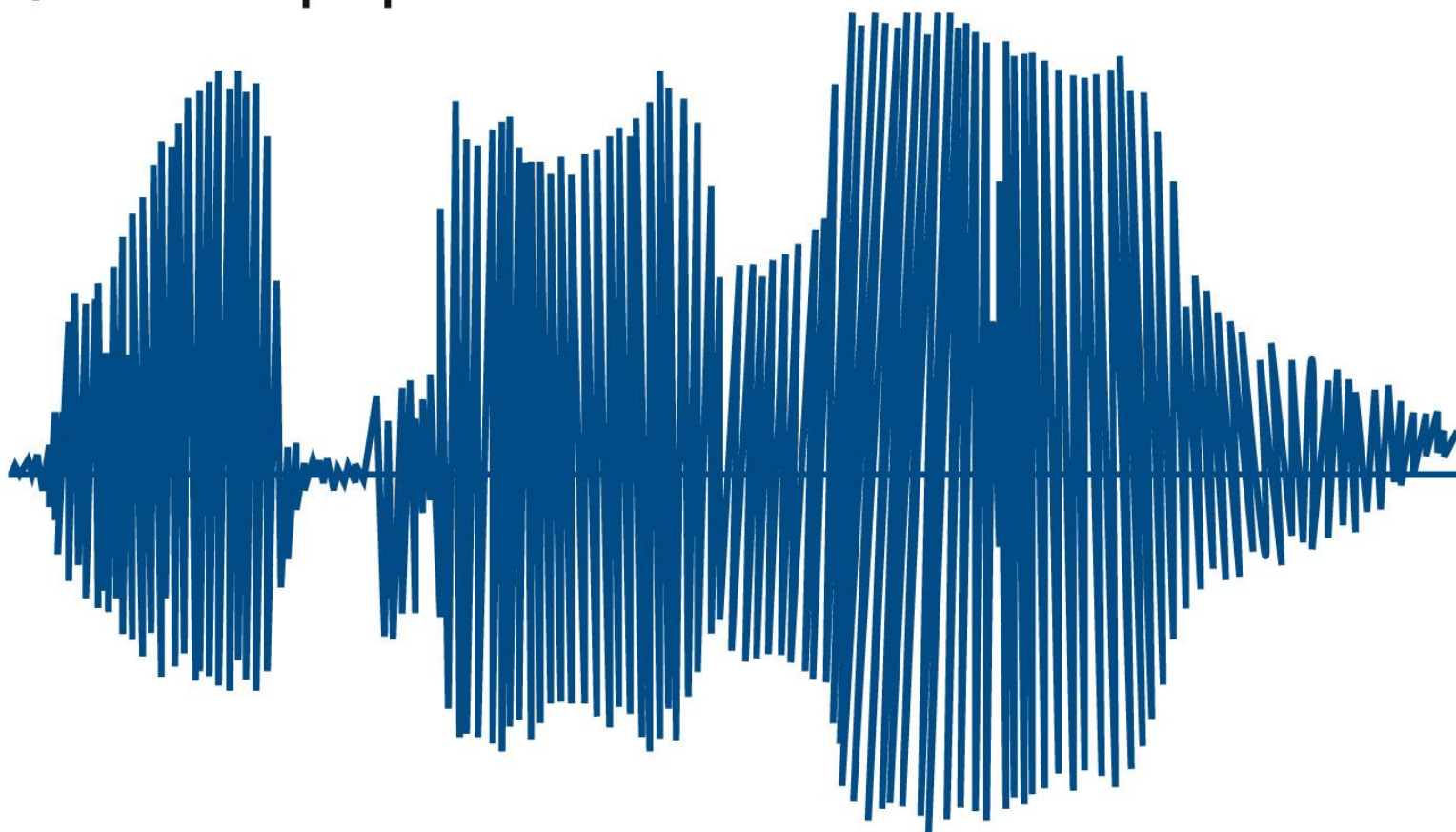
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Language

What

do you mean?



Language

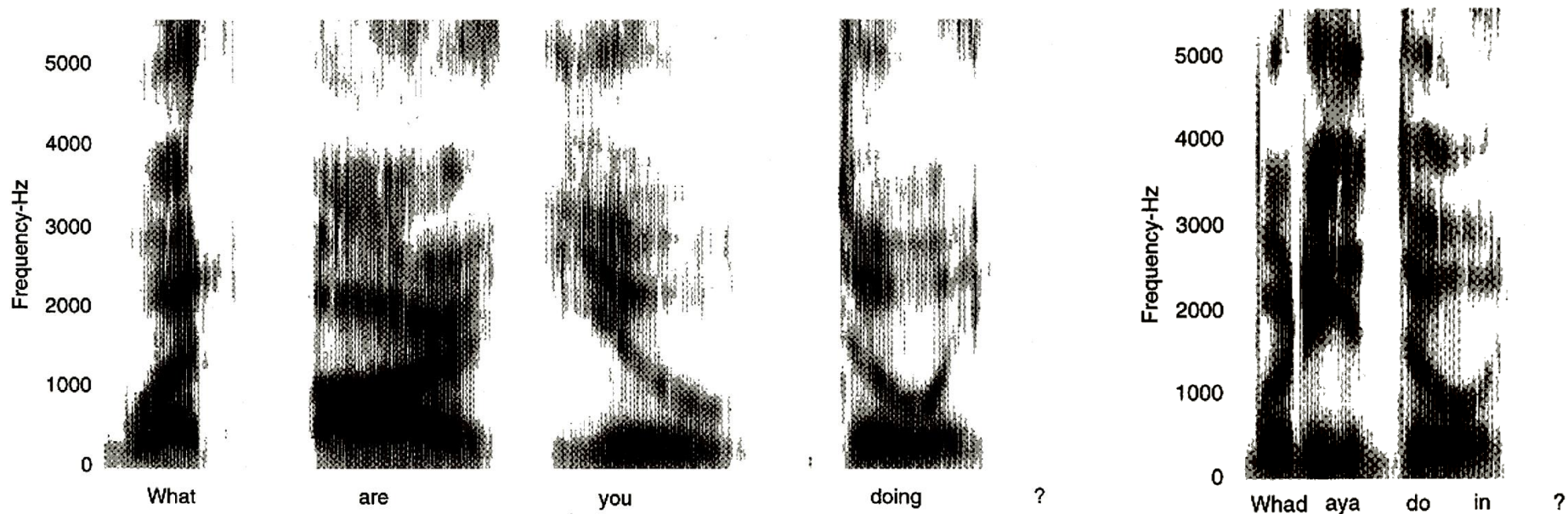
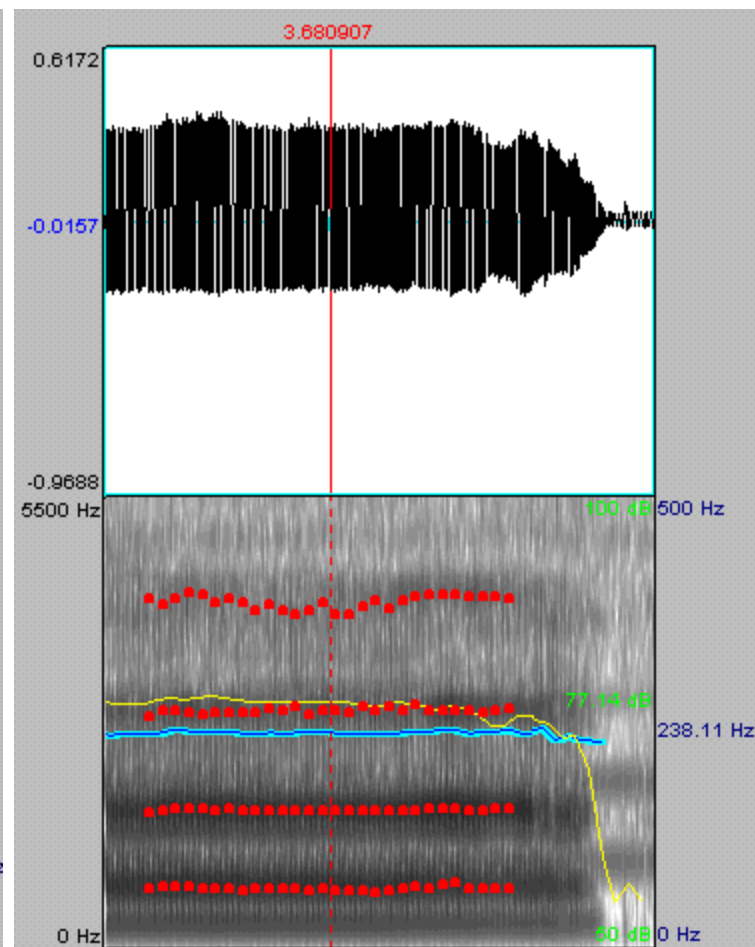
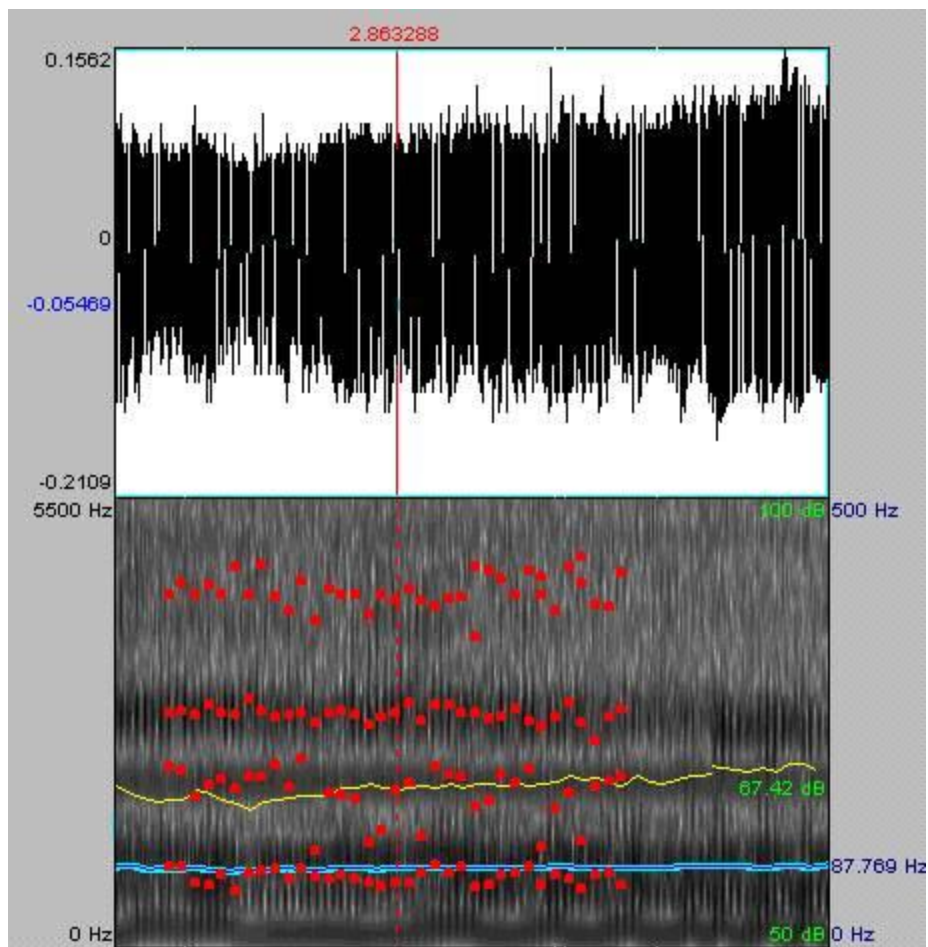


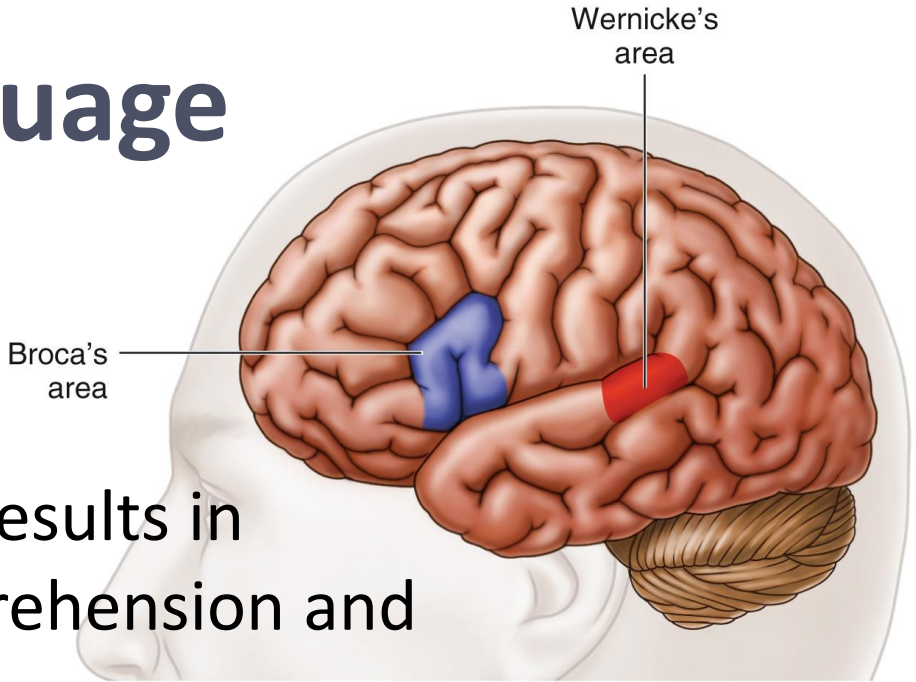
Figure 13.7

(a) Spectrogram of "What are you doing?" pronounced slowly and distinctly. (b) Spectrogram of "What are you doing?" as pronounced in conversational speech. (Spectrograms courtesy of David Pisoni.)

Language



Language



- **Aphasia:**

- a language disorder that results in deficits in language comprehension and production

- **Expressive aphasia (Broca's aphasia):**
 - interruption of ability to speak
- **Receptive aphasia (Wernicke's aphasia):**
 - difficulties to understand the meaning of words



TactusTherapy



TactusTherapy

Overview

- Mental representations
- Decision making
- Problem solving
- Language

